# $\gamma_5$ in the Drell–Yan computation

When is $aa = vv$? Larin's scheme, open lines and pure singlet

DY_P2B — notes_Larin/

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**Abstract**

The vector boson $V = \gamma^*, Z$ couples to quarks through $\gamma^\mu(v_q + a_q\gamma_5)$. Squaring gives vector–vector, axial–axial and mixed pieces. We explain when the axial–axial piece equals the vector–vector one ($aa = vv$) and when it does not, what Larin's prescription for $\gamma_5$ does in squared amplitudes, and which axial-current renormalisation constants are needed. The rule: $aa = vv$ when $V$ sits on the *same open quark line* in the amplitude and in the complex-conjugate amplitude, and only then. How to assemble all contributions with their counterterms: `assembly.pdf`.

## Summary

Labels of the $V$ vertex (as in `assembly.pdf`): $v$, $a$ on open quark line 1 (flavour $q$); $v_2$, $a_2$ on open quark line 2 (flavour $Q$); $v_{\rm PS}$, $a_{\rm PS}$ on a closed quark loop, the **pure singlet (PS)**, with couplings $\sum_f v_f$, $\sum_f a_f$.

| $V$ in amplitude / conjugate | example | axial–axial |
|---|---|---|
| same open line | $v\,v^*$, $v_2 v_2^*$ | $aa = vv$ (anticommuting $\gamma_5$) |
| different open lines | $v\,v_2^*$ | $aa_2 \neq vv_2$, also in $d = 4$: Larin |
| identical quarks ($q = Q$) | both kinds, same couplings | split by line |
| closed loop (PS) | $v_{\rm PS}v^*$, $a_{\rm PS}a^*$ | vector: compute; axial: Larin, $\propto \sum_f a_f$ |

The parity-odd pieces ($va$, $av$) contain one $\gamma_5$ in the hadronic trace and always need Larin (section 1).

## 1 The leptonic tensor

For massless leptons, $V \to \ell^-(l_1)\ell^+(l_2)$ with vertex $\gamma^\mu(v_\ell + a_\ell\gamma_5)$:

$$L^{\mu\nu} = \mathrm{Tr}\big[\not{l}_1\gamma^\mu(v_\ell + a_\ell\gamma_5)\not{l}_2\gamma^\nu(v_\ell + a_\ell\gamma_5)\big] = (v_\ell^2 + a_\ell^2)\,4\big(l_1^\mu l_2^\nu + l_1^\nu l_2^\mu - g^{\mu\nu} l_1\!\cdot\! l_2\big) + 2v_\ell a_\ell\,\mathrm{Tr}[\not{l}_1\gamma^\mu\not{l}_2\gamma^\nu\gamma_5]. \tag{1}$$

(The conjugate vertex is again $\gamma^\nu(v_\ell + a_\ell\gamma_5)$, since $\gamma^0(\gamma^\nu\gamma_5)^\dagger\gamma^0 = \gamma^\nu\gamma_5$; the first $(v_\ell + a_\ell\gamma_5)$ moves through $\not{l}_2\gamma^\nu$ without sign, and $(v_\ell + a_\ell\gamma_5)^2 = v_\ell^2 + a_\ell^2 + 2v_\ell a_\ell\gamma_5$.) The symmetric part multiplies the parity-even hadronic pieces ($vv$, $aa$, …), the antisymmetric ($\epsilon$) part, with coupling $2v_\ell a_\ell$, the parity-odd ones ($va$, $av$, …).

Averaged over the lepton angles at fixed $q = l_1 + l_2$: $\langle l_1^\rho l_2^\sigma\rangle = A\,q^\rho q^\sigma + B\,q^2 g^{\rho\sigma}$; contracting with $g_{\rho\sigma}$ and $q_\rho q_\sigma$ gives $q^2/2 = (A + 4B)q^2$, $q^4/4 = (A + B)q^4$, so $A = \frac{1}{6}$, $B = \frac{1}{12}$: the antisymmetric part averages to zero and $\langle L^{\mu\nu}\rangle = \frac{4}{3}(v_\ell^2 + a_\ell^2)(q^\mu q^\nu - q^2 g^{\mu\nu}) = \frac{4}{3}(v_\ell^2 + a_\ell^2)L_1^{\mu\nu}$. The parity-odd pieces contribute only to observables that resolve the lepton direction; there they are projected with the antisymmetric projector $L_6$.

## 2 $\gamma_5$ in $d$ dimensions; Larin's scheme

In $d \neq 4$ one cannot have both $\{\gamma_5, \gamma^\mu\} = 0$ and $\mathrm{Tr}[\gamma_5\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta] \propto \epsilon^{\alpha\beta\gamma\delta} \neq 0$ (anticommutation and cyclicity force that trace to vanish). Anticommuting $\gamma_5$ is therefore unambiguous only if no trace with an odd number of $\gamma_5$ has to be computed.

Larin [hep-ph/9302240] replaces the axial vertex by

$$\gamma^\mu\gamma_5 \;\to\; \tfrac{1}{2}(\gamma^\mu\gamma_5 - \gamma_5\gamma^\mu) = \tfrac{i}{6}\,\epsilon^{\mu\nu\rho\sigma}\gamma_\nu\gamma_\rho\gamma_\sigma \tag{2}$$

(normalisation of arXiv:2306.10170, eq. (4.3)) and contracts the $\epsilon$ tensors in $d$ dimensions. No $\gamma_5$ is ever moved, so the scheme is consistent; the axial current must then be renormalised by

$$Z_a^{ns} = 1 - 2C_F a + \Big(\tfrac{11}{2}C_F^2 - \tfrac{107}{36}C_F C_A + \tfrac{1}{18}C_F n_f + \frac{C_F\beta_0}{\epsilon}\Big)a^2, \tag{3}$$

$$Z_s = C_F T_R\Big(\frac{3}{2\epsilon} + \frac{3}{4}\Big)a^2 \quad \text{(singlet, per flavour)}, \tag{4}$$

with $a = \alpha_s/(2\pi)$, $\beta_0 = \frac{11}{6}C_A - \frac{2}{3}T_R n_f$. The finite parts restore the anticommuting-$\gamma_5$ Ward identities; the pole $C_F\beta_0/\epsilon$ is the $\overline{\mathrm{MS}}$ part. Conversion check (Larin's constants as I recall them, in $a_s = \alpha_s/4\pi = a/2$, to be checked against the original): $Z_5^{ns} = 1 - 4C_F a_s + a_s^2(22C_F^2 - \frac{107}{9}C_A C_F + \frac{2}{9}C_F n_f)$, $Z_{\overline{\mathrm{MS}}}^{ns} = 1 + a_s^2(\frac{22}{3}C_A C_F - \frac{4}{3}C_F n_f)/\epsilon$; with $a_s^2 = a^2/4$ these give the expression above. The singlet finite constant has $\frac{31}{18}C_F n_f$ instead of $\frac{2}{9}C_F n_f$: the difference $\frac{3}{2}C_F n_f a_s^2 = \frac{3}{4}C_F T_R n_f a^2$ is the $\frac{3}{4}$ of $Z_s$ (arXiv:2306.10170, eqs. (5.5), (5.6)).

*Remark on arXiv:2306.10170, eq. (5.13):* the $\bar{G}^{(0),n}$ coefficient is printed as $2\beta_0/\epsilon$; expanding $Z_a^{ns}\sqrt{\bar{a}}$ gives $2C_F\beta_0/\epsilon$ ($C_F\beta_0/\epsilon$ from $(-2C_F)(-\beta_0/2\epsilon)$ and $C_F\beta_0/\epsilon$ from the pole of $Z_a^{ns}$): probably a missing $C_F$.

## 3 Same line: $aa = vv$

One open massless quark line, $V$ attached once in the amplitude and once in the conjugate, any number of gluons. The spin-summed product is one trace,

$$\mathrm{Tr}\big[\not{p}\,\Gamma_1\,\gamma^\mu\gamma_5\,\Gamma_2\,\not{p}'\,\bar{\Gamma}_2'\,\gamma^\nu\gamma_5\,\bar{\Gamma}_1'\big], \tag{5}$$

$\Gamma_2$ = string from the $V$ vertex to the end of the line in the amplitude, $\bar{\Gamma}_2'$ the same in the conjugate (reversed). Along a massless line propagators $\not{k}$ and vertices $\gamma^\alpha$ alternate and the line ends with a vertex, so $\Gamma_2 = \not{k}_1\gamma^{\alpha_1}\cdots\not{k}_j\gamma^{\alpha_j}$ has $2j$ matrices; the same for $\bar{\Gamma}_2'$. Between the two $\gamma_5$ there are $2j + 1 + 2j' + 1$ matrices: an *even* number. With anticommuting $\gamma_5$, moving one $\gamma_5$ next to the other costs $+1$, and $\gamma_5^2 = 1$:

$$\boxed{\mathrm{Tr}[\cdots\gamma^\mu\gamma_5\cdots\gamma^\nu\gamma_5\cdots] = \mathrm{Tr}[\cdots\gamma^\mu\cdots\gamma^\nu\cdots] \quad\Rightarrow\quad aa = vv} \tag{6}$$

diagram by diagram, in any $d$, also with loops on the line (gluon loops, quark loops without $V$). No trace with a single $\gamma_5$ is computed: the two are removed before tracing. Helicity version: $P_\pm\gamma^\alpha = \gamma^\alpha P_\mp$, $P_\pm\not{k} = \not{k}P_\mp$, so chirality $h$ is conserved along the line, $A_a(h) = h\,A_v(h)$ and $\sum_h |A_a|^2 = \sum_h h^2|A_v|^2 = \sum_h |A_v|^2$. The parity-odd $\sum_h A_v A_a^* = \sum_h h|A_v|^2$ keeps one $\gamma_5$ in the trace: it needs Larin.

**Larin breaks it at $O(\epsilon)$.** Larin's vertex is not $\gamma^\mu\gamma_5$ in $d$ dimensions. Born:

$$vv = N_c(d-2)z^{-3}, \qquad aa_{\mathrm{Larin}} = -\tfrac{1}{6}N_c(d-2)(d-3)(d^2-8d+10)z^{-3}, \tag{7}$$

$$\frac{aa_{\mathrm{Larin}}}{vv} = (1-2\epsilon)\Big(1 - \tfrac{2}{3}\epsilon^2\Big). \tag{8}$$

Equal at $d = 4$, different at $O(\epsilon)$; the same pattern holds for the double-real same-line pieces. Larin $+Z_a^{ns}$ and anticommuting $\gamma_5$ are both consistent and give the same physical non-singlet result; the $O(\epsilon)$ differences cancel only in the full combination (reals, virtuals, collinear counterterms in the same scheme).

**Choice.** Same-line parity-even pieces: anticommuting $\gamma_5$, i.e. compute $vv$ and set $aa = vv$, at all orders (RRR, RR, R, B and the non-singlet part of the loop contributions). Then $Z_a^{ns}$ is never needed for them. Everything else with an axial vertex (cross-line, PS, parity-odd): Larin $+Z$'s.

## 4 Different lines: $aa \neq vv$

$q\bar{q} \to VQ\bar{Q}$ with $V$ on the $q$ line in the amplitude and on the $Q$ line in the conjugate: for $q \neq Q$ the product is two traces with one $\gamma_5$ each, and the two $\gamma_5$ cannot meet. With chiralities $h_1$, $h_2$ of the lines,

$$\sum_{h_1 h_2} A_a^{(1)} A_a^{(2)*} = \sum_{h_1 h_2} h_1 h_2 \, A_v^{(1)} A_v^{(2)*} \neq \sum_{h_1 h_2} A_v^{(1)} A_v^{(2)*} : \tag{9}$$

configurations with $h_1 = -h_2$ enter with opposite sign, so $aa_2 \neq vv_2$ already in four dimensions. Each trace has a single $\gamma_5$, so anticommuting $\gamma_5$ is ambiguous: these pieces need Larin (and $Z_a^{ns}$ on both vertices in their loop corrections). Their coupling is $a_q a_Q$; summed over $Q$, $\sum_Q a_Q$ cancels within complete generations except for the $b$.

**Identical quarks** ($q\bar{q} \to Vq\bar{q}$): both lines have the same couplings, so $v_q^2$ and $a_q^2$ multiply same-line products, cross-line products and the exchange interference together. $aa = vv$ holds only for the same-line ones: the pieces must be separated by line, not by coupling.

## 5 Closed quark loops: pure singlet

$V$ on a closed loop, summed over the $n_f$ flavours of the loop: couplings $\sum_f v_f$, $\sum_f a_f$; only in loop amplitudes.

- **Vector PS.** Furry: zero with two gluons on the loop ($V$ + 1 parton at one loop; four-quark processes at one loop; $q\bar{q} \to V$ at two loops). With three gluons on the loop ($d^{abc}$) it is non-zero: $V$ + 1 parton at two loops, $q\bar{q} \to Vgg$ at one loop, $gg \to Vg$ at one loop. It does not cancel between flavours ($v_u \neq -v_d$).

- **Axial PS.** One $\gamma_5$ in the loop: the anomalous triangle; Larin $+Z_s$ is required. Example: the one-loop $a_{\rm PS}$ amplitude for $q\bar{q} \to Vg$ times the tree $a^*$ (an NNLO RV piece) is non-zero by itself (Landau–Yang does not apply: one gluon is off shell); its coupling $a_q \sum_f a_f$ makes it vanish, together with its $Z_s$ counterterm, in the sum over a complete mass-degenerate doublet ($a_u = -a_d$).

- With $n_f = 5$ massless flavours the $b$ is left without its partner: $\sum_f a_f \neq 0$. A correct treatment of this remainder needs the massive top in the loop (arXiv:2306.10170, introduction); in DY it is usually small and treated separately. To check in the N$^3$LO DY literature before deciding.

- $gg \to Vg$: vector $\propto d^{abc}$, axial $\propto f^{abc}$, so their interference (parity-odd) vanishes by colour.

- $\gamma^*$: $a_f = 0$. $W$: no PS (a $W$ cannot couple to a loop that connects only to gluons).